MATH 131AH: Week 1 (September 25 – October 2, 2026)

Notes from the first week of honors analysis: logical statements and quantifiers, the Peano axioms and induction, equivalence relations and the construction of $\mathbb{Q}$, fields and ordered fields, upper bounds and suprema, the least upper bound property, the Archimedean property, the density of $\mathbb{Q}$ in $\mathbb{R}$, and a discussion section on proof techniques.

1 Lecture, September 25: Logic and the Peano Axioms

Reading: Visan Lec. 1 – Lec. 2.

1.1 Logical Statements

Let A, B be statements. A denotes "A is true". "If A, then B" means $A \Rightarrow B$.

[Sketch: a circle A drawn inside a larger circle B.]

A if and only if B: $A = B$.

Negation.

$$\sim (A \Rightarrow B) = [A \cap \sim B \neq \emptyset] \tag{1}$$

1.2 Quantifiers

$\forall$: "for any"; $\exists$: "there exists".

Example 1.1. Let $X, Y \subseteq \mathbb{R}$.

1. $\sim (\forall x \in X,\ f(x) = 0) \Rightarrow \exists\, x_0 \in X$ s.t. $f(x_0) \neq 0$.
2. $\sim [\exists\, y \in Y,\ f(y) = 0] \Rightarrow \forall y \in Y,\ f(y) \neq 0$.
3. $\sim (\exists\, A, B \in X$ s.t. $A \leq y \leq B$ for $\forall y \in Y) = \forall A, B \in X,\ \exists\, y \in Y$ s.t. $y < A$ or $B < y$.
4. $\sim (\exists\, M > 0$ s.t. $|f(x) - f(y)| < M|x - y|\ \forall x, y \in H) = \forall M > 0$, there exists some $x_0, y_0 \in H$ s.t. $|f(x_0) - f(y_0)| \geq M|x_0 - y_0|$.

In math, sets are often defined by their properties (axioms), giving structures and abstractions, so that we understand the properties of the number system we use.

1.3 The Peano Axioms

Definition 1.2. $\mathbb{N}$ is the number system that satisfies the *Peano axioms*:

1. $1 \in \mathbb{N}$.
2. $\exists$ a successor function $S\colon \mathbb{N} \to \mathbb{N}$ s.t. it is 1-1, "$S(n) = n + 1$".
3. $\nexists\, x \in \mathbb{N}$ s.t. $S(x) = 1$.

4. If $S \subseteq \mathbb{N}$ and if S satisfies 1 and 2, then $S = \mathbb{N}$.

$\to \mathbb{N}$ is the smallest set with properties 1 and 2.

Example 1.3 (Induction)**.** Show that $2^{n-1} \leq n!$ for all $n \in \mathbb{N}$.

1. True if $n = 1$: $2^0 \leq 1!$.
2. Assume true for $n = k$. We need to prove $2^{k+1-1} \leq (k+1)!$. Using the case $n = k$ and $k \geq 1$,

$$2^k = 2^{k-1} \cdot 2 \tag{2}$$
$$\leq k! \cdot 2 \leq k!\,(k+1) = (k+1)! \tag{3}$$

Let $S = \{k \in \mathbb{N} \text{ s.t. } 2^{k-1} \leq k!\}$. S satisfies 1 and 2 ($S(n) = n + 1$), so by 4, $S = \mathbb{N}$.

2 Lecture, September 28: Equivalence Relations and the Construction of $\mathbb{Q}$

Last time we talked about characterizing $\mathbb{N}$ with Peano's axioms:

$$\mathbb{N} = \{1, S(1), S(S(1)), \dots\} \tag{4}$$

where $S\colon \mathbb{N} \to \mathbb{N}$, $S(1) = 2$, $S(2) = 3$, $S(k-1) = k$. The proof of the existence of $\mathbb{N}$ involves set theory (skip). We accept $\mathbb{N}$, $+$, $\cdot$, and

$$\mathbb{Z} = \mathbb{N} \cup (-\mathbb{N}) \cup \{0\}. \tag{5}$$

Constructing $\mathbb{Q}$ in Lec. 3.

2.1 Equivalence Relations

Definition 2.1 (Equivalence relation)**.** $\sim$ is an equivalence relation on a set A if:

1. $x \sim x$ for all $x \in A$;
2. $\forall x, y \in A$, if $x \sim y$ then $y \sim x$;
3. $\forall x, y, z \in A$, if $x \sim y$ and $y \sim z$, then $x \sim z$.

Example 2.2. $A = \mathbb{N}$.

1. $x \sim y$ if $x = y$.
2. (mod 3) $x \sim y$ if $x - y = 3k$ for $k \in \mathbb{Z}$. If $k \in \mathbb{N}$, property 2 breaks.

Definition 2.3. Given a set A and an equivalence relation $\sim$ on A, we define the equivalence class of $a \in A$ by

$$C(a) := \{b \in A : a \sim b\}. \tag{6}$$

1. Either $C(a) = C(b)$ or $C(a) \cap C(b) = \emptyset$ (from property 3).

[Sketch: two overlapping classes $C(a)$, $C(b)$, with $x \in C(a)$, $y \in C(b)$ and z in the overlap.] If $\exists\, z \in C(a) \cap C(b)$, then for all $x \in C(a)$ and $y \in C(b)$, $x \sim z$ and $z \sim y \Rightarrow x \sim y$.

Example 2.4. For the mod 3 relation, $\mathbb{N} = C(1) \cup C(2) \cup C(3)$.

The set of equivalence classes is

$$A/\sim := \{C(a); a \in A\}. \tag{7}$$

2.2 Defining $\mathbb{Q}$

Definition 2.5. Let

$$F = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : b \neq 0\}, \tag{8}$$
$$\mathbb{Q} := F/\sim, \tag{9}$$

where $(a, b) \sim (c, d)$ if $ad = bc$. Denote $\frac{a}{b} := C((a, b))$.

Remark. Homework: check $\sim$ is an equivalence relation.

1. $(a, b) \sim (a, b)$.
2. $(a, b) \sim (c, d) \Rightarrow (c, d) \sim (a, b)$: $ad = bc \to cb = da$.
3. (i) $(a, b) \sim (c, d)$ and (ii) $(c, d) \sim (e, f)$. WTS $(a, b) \sim (e, f)$, i.e. $af = be$. We have

$$\text{(i) } ad = bc, \qquad \text{(ii) } cf = de, \tag{10}$$
$$\text{(ii)} \times b\text{:} \quad cfb = deb. \tag{11}$$

Then, write

$$adf = bcf \tag{12}$$
$$= deb, \tag{13}$$

so $af = be$ since $d \neq 0$, by definition of the set F.

Operations on $\mathbb{Q}$: $+$, $\cdot$

$$\frac{a}{b} + \frac{c}{d} := \frac{ad + bc}{bd} \tag{14}$$
$$\frac{a}{b} \cdot \frac{c}{d} := \frac{ac}{bd} \tag{15}$$

If $a_1/b_1 = a_2/b_2$ and $c_1/d_1 = c_2/d_2$, then

$$\frac{a_1}{b_1} + \frac{c_1}{d_1} = \frac{a_2}{b_2} + \frac{c_2}{d_2}. \tag{16}$$

2.3 The Square Root of 2

Lemma 2.6. *There is no $x \in \mathbb{Q}$ s.t. $x^2 = 2 \iff \forall x \in \mathbb{Q}$, $x^2 \neq 2$.*

Proof. Negate for proof by contradiction: suppose $\exists\, x_0 \in \mathbb{Q}$ s.t. $x_0^2 = 2$. Then $x_0 = \frac{m}{n}$, $m, n \in \mathbb{Z}$, $n \neq 0$, with m, n mutually prime. Hence,

$$m^2 = 2n^2, \tag{17}$$

so m is even: $m = 2k$, $k \in \mathbb{Z}$. Then

$$(2k)^2 = 4k^2 = 2n^2, \tag{18}$$

so n is even, a contradiction. □

3 Lecture, September 30: Fields and Ordered Fields

Last time: we constructed $\mathbb{Q}$ via equivalence classes and defined $+$, $\cdot$ on $\mathbb{Q}$.

3.1 Fields

Definition 3.1. A field is a set F with operations $+$, $\cdot$, written $(F, +, \cdot)$, s.t.:

1. $x, y \in F \to xy,\ x + y \in F$;
2. $x + y = y + x, \quad xy = yx$;
3. $(x + y) + z = x + (y + z), \quad (xy)z = x(yz)$;
4. $\exists\, 0 \in F$ s.t. $x + 0 = x$, $\forall x \in F$;
5. $\exists\, 1 \in F$ s.t. $1 \cdot x = x$, $\forall x \in F$, $1 \neq 0$;
6. $\exists\, (-x) \in F$ s.t. $x + (-x) = 0$, $\forall x \in F$;
7. $\forall x \in F$, $x \neq 0$, $\exists\, \frac{1}{x} \in F$ s.t. $x \cdot \frac{1}{x} = 1$;
8. $x(y + z) = xy + xz$.

Example 3.2. $(\mathbb{Q}, +, \cdot)$ is a field (homework). Here $0 = \frac{0}{1}$ and $1 = \frac{1}{1}$. For $x \in \mathbb{Q}$, if $x \neq 0$, $x = \frac{a}{b}$, then $\frac{1}{x} = \frac{b}{a}$. It is well defined because $\frac{a}{b} \sim \frac{c}{d} \to \frac{b}{a} \sim \frac{d}{c}$, since $ad = bc$.

Proposition 3.3. *Let $(F, +, \cdot)$ be a field. Then the following hold.*

1. *$\forall x, y, z \in F$:*
   - *(a) $x + y = x + z \Rightarrow y = z$;*
   - *(b) $x + y = x \to y = 0$;*
   - *(c) $x + y = 0 \to y = -x$.*
2. *$\forall x, y, z \in F$, $x \neq 0$:*
   - *(a) $xy = xz \to y = z$;*
   - *(b) $xy = x \to y = 1$;*
   - *(c) $xy = 1 \to y = \frac{1}{x}$.*

Proof. 1(a): adding $-x$ to both sides,

$$(-x) + (x + y) = (-x) + (x + z) \tag{19}$$
$$((-x) + x) + y = ((-x) + x) + z \tag{20}$$
$$0 + y = 0 + z \tag{21}$$
$$y = z. \tag{22}$$

1(b):

$$x + y = x + 0 \tag{23}$$
$$(-x) + (x + y) = (-x) + (x + 0) \tag{24}$$
$$((-x) + x) + y = ((-x) + x) + 0 \tag{25}$$
$$0 + y = 0 + 0 \Rightarrow y = 0. \tag{26}$$

1(c):

$$(-x) + (x + y) = -x \tag{27}$$
$$((-x) + x) + y = -x \tag{28}$$
$$0 + y = -x \tag{29}$$
$$y = -x. \tag{30}$$

2(a): multiplying by $\frac{1}{x}$,

$$\tfrac{1}{x}(xy) = \tfrac{1}{x}(xz) \tag{31}$$
$$(\tfrac{1}{x} \cdot x)y = (\tfrac{1}{x} \cdot x)z \tag{32}$$
$$1 \cdot y = 1 \cdot z \tag{33}$$
$$y = z. \tag{34}$$

2(b):

$$\tfrac{1}{x}(xy) = \tfrac{1}{x}(x) \tag{35}$$
$$(\tfrac{1}{x} \cdot x)y = (\tfrac{1}{x} \cdot x) \tag{36}$$
$$1 \cdot y = 1 \tag{37}$$
$$y = 1. \tag{38}$$

2(c):

$$\tfrac{1}{x}(xy) = \tfrac{1}{x} \cdot 1 \tag{39}$$
$$(\tfrac{1}{x} \cdot x)y = \tfrac{1}{x} \tag{40}$$
$$1 \cdot y = \tfrac{1}{x} \tag{41}$$
$$y = \tfrac{1}{x}. \tag{42}$$

□

Proposition 3.4. *$\forall x, y, z \in F$:*

1. *$0 \cdot x = 0$;*
2. *$x \neq 0$, $y \neq 0 \to x \cdot y \neq 0$;*

3. $(-x) \cdot y = -(xy) = x(-y)$;

4. $(-x)(-y) = xy$.

Proof. 1: write

$$0 \cdot x = (0+0)x = 0 \cdot x + 0 \cdot x, \tag{43}$$
$$0 \cdot x = 0 + 0 \cdot x, \tag{44}$$

hence $0 \cdot x + 0 \cdot x = 0 + 0 \cdot x$, and by 1(a), $0 \cdot x = 0$.

2: if $x \cdot y = 0 = x \cdot 0$, then by 2(a), $y = 0$, a contradiction.

3, 4: check. □

3.2 Orders

Definition 3.5. An order $<$ is a relation on a set S with the following properties:

1. if $x, y \in S$, then either $x < y$ or $x = y$ or $y < x$;
2. if $x, y, z \in S$ and $x < y$ and $y < z$, then $x < z$.

$(S, <)$ is an ordered set if an order $<$ is defined on S.

Example 3.6.

1. $\mathbb{N}$: $a < b$ if b is an offspring of a: $b = S \circ S \circ \cdots \circ S(a)$.
2. $\mathbb{Z}$: $a < b$ if $b - a \in \mathbb{N}$.
3. $\mathbb{Q}$: for $x, y \in \mathbb{Q}$, $x = \frac{a}{b}$, $y = \frac{c}{d}$ with $b, d > 0$, $bc - ad > 0 \to x < y$.
4. $\mathbb{C} = \{a + ib; a, b \in \mathbb{R}\}$: $a + bi < c + di$ if either $a < c$, or $a = c$ and $b < d$.

3.3 Ordered Fields

Definition 3.7. A set F, $(F, +, \cdot, <)$, is an ordered field if $(F, +, \cdot)$ is a field and $(F, <)$ is an ordered set s.t.

1. $y < z \Rightarrow x + y < x + z$;
2. $x > 0$, $y > 0 \Rightarrow xy > 0$.

Example 3.8. $(\mathbb{Q}, +, \cdot, <)$.

Next time: $\mathbb{R}$ will be characterized.

4 Lecture, October 2: Ordered Fields, Bounds, and the Real Numbers

Last time: $(F, +, \cdot)$: field; $(S, <)$: ordered set.

4.1 $\mathbb{Q}$ Is an Ordered Field

Example 4.1. $\mathbb{Q}$ is an ordered field. (Recall $\frac{a}{b} < \frac{c}{d}$ if $b, d > 0$ and $bc - ad > 0$; check Visan Lec. [**illegible**], Thm. 5.2.)

Let us check that the order $<$ is well defined: if $(a, b) \sim (a_1, b_1)$ and $(c, d) \sim (c_1, d_1)$, then

$$\frac{a}{b} < \frac{c}{d} \to \frac{a_1}{b_1} < \frac{c_1}{d_1}. \tag{45}$$

Proof. We have $bc - ad > 0$. WTS $b_1c_1 - a_1d_1 > 0$. We have

$$(1)\ ab_1 = ba_1, \qquad (2)\ cd_1 = dc_1. \tag{46}$$

Then, write

$$bc > ad \tag{47}$$
$$b_1bc > b_1ad \tag{48}$$
$$b_1bc > ba_1d \quad \Rightarrow \quad b_1c > a_1d \qquad (b > 0) \tag{49}$$
$$d_1b_1c > d_1a_1d \tag{50}$$
$$b_1c_1d > d_1a_1d \quad \Rightarrow \quad b_1c_1 > d_1a_1 \qquad (d > 0). \tag{51}$$

□

4.2 Properties of Ordered Fields

Proposition 4.2. *Let $(F, +, \cdot, <)$ be an ordered field.*

(a) $x > 0 \iff -x < 0$.

(b) $x > 0$, $y < z \Rightarrow xy < xz$; $\quad x < 0$, $y < z \Rightarrow xy > xz$.

(c) $x \neq 0 \to x^2 > 0$.

(d) $0 < x < y \to 0 < \frac{1}{y} < \frac{1}{x}$.

Proof. (a): Let $x > 0$. Then

$$x + (-x) > 0 + (-x) \tag{52}$$
$$0 > -x. \tag{53}$$

(b): Let $x > 0$, $y < z$. Then

$$z - y > 0 \tag{54}$$
$$\Rightarrow x(z - y) > 0. \tag{55}$$

If $x < 0$ and $y < z$, then $-x > 0$ and $z - y > 0$, so

$$(-x)(z - y) > 0 \tag{56}$$
$$-(x(z - y)) > 0 \tag{57}$$
$$x(z - y) < 0. \tag{58}$$

(c): $x \neq 0 \to x > 0$ or $x < 0$. If $x > 0$: $x \cdot x > 0$. If $x < 0$: $(-x) \cdot (-x) = x \cdot x > 0$.

(d): Let $x > 0$. If $\frac{1}{x} \leq 0$, then $-\frac{1}{x} \geq 0$, and

$$x \cdot \left(-\tfrac{1}{x}\right) \geq 0 \tag{59}$$

$$x \cdot \left(-\tfrac{1}{x}\right) = -1 \geq 0, \tag{60}$$

a contradiction. So we have $x > 0$, $y > 0 \to \frac{1}{x}, \frac{1}{y} > 0$. With $x < y$,

$$\left(\tfrac{1}{x} \cdot \tfrac{1}{y}\right) x < \left(\tfrac{1}{x} \cdot \tfrac{1}{y}\right) y \tag{61}$$

$$\tfrac{1}{y} < \tfrac{1}{x}. \tag{62}$$

□

4.3 Bounds, Maximum and Minimum

Definition 4.3. Let $(S, <)$ be an ordered set and let $E \subseteq S$. If $\exists\, M \in S$ s.t. $x \leq M$ for all $x \in E$, we say that M is an upper bound of E, and "E is bounded above". If $\exists\, m \in S$ s.t. $m \leq x$ for all $x \in E$, then m is a lower bound of E.

If $M \in E$, we call $M = \max(E)$. If $m \in E$, $m = \min(E)$.

Example 4.4. $E = \{x \in \mathbb{Q},\ x > 0 : x^2 \leq 4\}$. $x = 2$ is an upper bound, because if $x > 2$, then

$$x \cdot x > 2 \cdot x > 2 \cdot 2 = 4, \tag{63}$$

so $x \notin E$. $x = 0$ is a lower bound, but $0 \notin E$, so $0 \neq \min E$. Here $\sup E = 2$ and $\inf E = 0$.

4.4 Supremum and Infimum

Definition 4.5. Suppose $E \subseteq S$ and $\alpha \in S$ satisfies:

1. α is an upper bound of E;
2. if M is another upper bound of E, then $\alpha \leq M$.

Then $\alpha = \sup E$ (supremum of E). Replacing "upper" by "lower" (and $\leq$ by $\geq$) gives $\beta = \inf E$ (infimum of E).

Example 4.6. $S = \mathbb{Q}$, $E = \{x \in \mathbb{Q},\ x^2 < 2\}$. An upper bound of E: 2. A lower bound of E: -2. There is no $\sup E$ and no $\inf E$ in $\mathbb{Q}$ ($\because$ Lem. 1.6).

Definition 4.7. $(S, <)$ has the *least upper bound property* if every non-empty $E \subseteq S$ that has an upper bound has a sup (l.u.b.). Similarly for inf (g.l.b.).

Let $E \subseteq S$, $(S, <)$. Recall $\alpha = \sup E$ if

1. α is an upper bound of E;
2. if β is an upper bound of E, then $\alpha \leq \beta$.

If $\exists \max E$, then $\max E = \sup E$: let $\alpha = \max E$, $\alpha \in E$.

1. α is an upper bound of E by definition.
2. If β is an upper bound of E, then $\alpha \leq \beta$ (since $\alpha \in E$).

Theorem 4.8. *If $(S, <)$ has the l.u.b. property, then it has the g.l.b. property.*

Proof. Let $E \subseteq S$, $E \neq \emptyset$. WTS: if E has a lower bound, then $\exists\, \alpha = \inf E$.

[Sketch: the set E on a line, with its lower bounds to the left.]

Let $L := \{\beta : \beta \text{ is a lower bound of } E\}$. $L \neq \emptyset$ (E has a lower bound), and for all $x \in E$, x is an upper bound of L. So by the l.u.b. property, $\exists\, \alpha = \sup L$.

Claim: $\alpha = \inf E$.

1. $\forall x \in E$, x is an upper bound of L, so $\alpha \leq x \to \alpha$ is a lower bound of E.
2. Let β be a lower bound of E; then $\beta \in L$ and α is an upper bound of L, so $\beta \leq \alpha$.

□

Remark. H.W. (marked next to the last step).

4.5 Characterizing $\mathbb{R}$

We are finally ready to characterize $\mathbb{R}$.

Theorem 4.9. *There exists $(\mathbb{R}, <, +, \cdot)$ that is an ordered field s.t.*

1. *$\mathbb{Q} \subseteq \mathbb{R}$ and $(\mathbb{Q}, <|_{\mathbb{Q}}, +|_{\mathbb{Q}}, \cdot|_{\mathbb{Q}})$ is an ordered field;*
2. *$\mathbb{R}$ has the l.u.b. property.*

Before constructing $\mathbb{R}$, we first discuss what properties of $\mathbb{R}$ 1–2 yield (suppose $\mathbb{R}$ exists for now).

Lemma 4.10 (Archimedean property)**.** *$\forall x, y \in \mathbb{R}$ with $x > 0$, $\exists\, n \in \mathbb{N}$ s.t. $nx > y$.*

Proof. Let $A := \{nx,\ n \in \mathbb{N}\}$. If the lemma is false, then y is an upper bound of A. By the l.u.b. property, $\exists\, \alpha := \sup A$. Since $x > 0$, $\alpha - x < \alpha$ is not an upper bound of A, so

$$\exists\, mx \in A \text{ s.t. } \alpha - x < mx \tag{64}$$
$$\Rightarrow \alpha < (m+1)x \in A, \tag{65}$$

a contradiction. □

4.6 Density of $\mathbb{Q}$

Definition 4.11. We say $A \subseteq \mathbb{R}$ is *dense* in $\mathbb{R}$ if for all $x, y \in \mathbb{R}$ s.t. $x < y$, $\exists\, a \in A$ s.t. $x < a < y$.

Lemma 4.12. *$\mathbb{Q}$ is dense in $\mathbb{R}$.*

Proof. Pick $x, y \in \mathbb{R}$, $x < y \Rightarrow (y - x) > 0$.

[Sketch: $x < y$ on a line; scaled up, nx and ny are more than 1 apart.]

By the Archimedean property, $\exists\, n \in \mathbb{N}$ s.t.

$$n(y - x) > 1. \tag{66}$$

$\exists\, m_1, m_2 \in \mathbb{N}$ s.t. $-m_2 < nx < m_1$ (why is this step necessary? To ensure that m is in a finite set), so

$$\exists\, m \in \mathbb{Z} \text{ s.t. } m - 1 \leq nx < m. \tag{67}$$

Now

$$m \leq nx + 1 < ny, \tag{68}$$

so $x < a = \frac{m}{n} < y$. □

Remark. HWK: show $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$.

4.7 Sequences

Definition 4.13. A sequence in $\mathbb{R}$ is a function $\mathbb{N} \to \mathbb{R}$, $n \mapsto a_n$, written $a = (a_n)_{n=1}^{\infty}$.

5 Discussion Section (Undated): Proof Techniques

Proofs: given a statement A, prove statement B.

5.1 Direct Proof

$A \longrightarrow B$.

Example 5.1. A: let $a, b \in \mathbb{R}$. B: $ab \leq a^2/2 + b^2/2$. Equivalently,

$$2ab \leq a^2 + b^2 \tag{69}$$
$$(a - b)^2 \geq 0. \tag{70}$$

Analysis tip: add and subtract; multiply and divide.

Example 5.2. Prove: if $a, b \in \mathbb{R}$, pick $\varepsilon > 0$; there exists $C(\varepsilon)$ s.t.

$$ab \leq \varepsilon a^2 + C(\varepsilon) b^2. \tag{71}$$

Write $ab = a'b'$ with $a' = a\sqrt{2\varepsilon}$ and $b' = \frac{b}{\sqrt{2\varepsilon}}$; then, by the previous example,

$$ab = a\sqrt{2\varepsilon} \cdot \frac{b}{\sqrt{2\varepsilon}} \tag{72}$$
$$\leq \varepsilon a^2 + \frac{b^2}{4\varepsilon}. \tag{73}$$

5.2 Induction

Induction: a sequence of statements A_n, $n = 1, 2, \ldots$

1. Step 1: prove $A(1)$.
2. Step 2: prove if $A(n)$ is true $\Rightarrow A(n+1)$ is true.

Example 5.3. Find all $n \in \mathbb{N}$ s.t. $2^n > n^3$. For the inductive step,

$$2^{n+1} = 2 \cdot 2^n > 2n^3 \geq (n+1)^3. \tag{74}$$

Find all n's s.t. $2n^3 \geq (n+1)^3$:

$$2 \geq \left(1 + \tfrac{1}{n}\right)^3 \tag{75}$$

$$2^{1/3} \approx 1.2599 \tag{76}$$

$$n \geq 4. \tag{77}$$

Check the first $A(n)$ that is true for $n \geq 4 \rightarrow$ Ans. $n = 10$.

5.3 Proof by Contradiction

Proof by contradiction of $A \Rightarrow B$.

Example 5.4. For all continuous $f\colon [0,1] \to [0,1]$, there exists $x^* \in [0,1]$ s.t. $f(x^*) = x^*$:

$$(\forall f\colon [0,1] \to [0,1] \text{ and cont.}),\ (\exists\, x^* \in [0,1]),\ (f(x^*) = x^*). \tag{78}$$

Fact: $\neg\forall = \exists$, $\neg\exists = \forall$. Negation:

$$(\exists f\colon [0,1] \to [0,1] \text{ cont.}),\ (\forall x \in [0,1]),\ (f(x) \neq x). \tag{79}$$

5.4 A Limit Problem

Example 5.5. Prove: if $f\colon [0,1] \to \mathbb{R}$ is continuous, then

$$\lim_{h \to 0^+} \frac{1}{\log h} \int_0^1 \frac{f(x)}{x+h}\, dx = -f(0). \tag{80}$$

Step 1: try stupid cases. If $f(x) = f(0)$,

$$\frac{1}{\log h} \int_0^1 \frac{f(0)}{x+h}\, dx \tag{81}$$

$$= f(0) \cdot \frac{\log(1+h)}{\log h} - f(0). \tag{82}$$

Step 2: reduction; assume $f(0) = 0$:

$$\frac{1}{\log h} \int_0^1 \frac{f(x)}{x+h}\, dx = \frac{1}{\log h} \int_0^1 \frac{f(x) - f(0)}{x+h}\, dx + \frac{1}{\log h} \int_0^1 \frac{f(0)}{x+h}\, dx. \tag{83}$$

Step 3: intuition. Pick δ small:

$$\frac{1}{\log h}\int_0^1 \frac{f(x)}{x+h}\,dx = \underbrace{\frac{1}{\log h}\int_0^\delta \frac{f(x)}{x+h}\,dx}_{\text{I}} + \underbrace{\frac{1}{\log h}\int_\delta^1 \frac{f(x)}{x+h}\,dx}_{\text{II}}. \tag{84}$$

I: small, since $f(0) = 0$. II:

$$\int_\delta^1 \frac{f(x)}{x+h}\,dx \le \max_{[0,1]} |f| \int_\delta^1 \frac{1}{x+h}\cdot\frac{1}{\log h}\cdots \tag{85}$$

[The last line runs off the edge of the page.]